

Computer Architecture

Term Test 2

30 Minutes

10 Marks

1. Answer the following short questions:

2 x 2 = 4

- Differentiate between spatial and temporal locality of reference.
- What advantages do fully associative memory address offer compared to directly mapped addressing?

2. Consider the following table, which shows the state of a 8-block cache where each block is 1 byte, for a upper-level memory which has 32 blocks. Now show the final state of the cache after the following operations are performed on it, along with explanations.

2 x 3 = 6

Index	V	Tag	Data
000	Y	10 _{two}	Memory (10000 _{two})
001	N		
010	Y	11 _{two}	Memory (11010 _{two})
011	N		
100	N		
101	N		
110	Y	10 _{two}	Memory (10110 _{two})
111	N		

- After trying to read address (00011₂)
- After trying to read address (10010₂)
- After trying to read address (00011₂) again